AUTOCARE

Project scope

V2(unfinished)

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AutoCare is an android based app, that lets users track everything about their personal vehicle such as upcoming required maintenance, maintenance history records, fuel consumption, driving habits (speed, parking, routes). The app will also allow users to find where they priviously parked their car.

1. TECHNOLOGY STACK

Component	Technology
Development environment	Android studio
Programing language	Kotlin
User interface	Jetpack Compose (Material 3)
Architecture	MVVM
Database(local)	Room (SQLite)
Device sensors	GPS, magnetometer, accelerometer
Location API	Undetermined yet
Map integration	Google Maps SDK
Version control	Git + GitHub

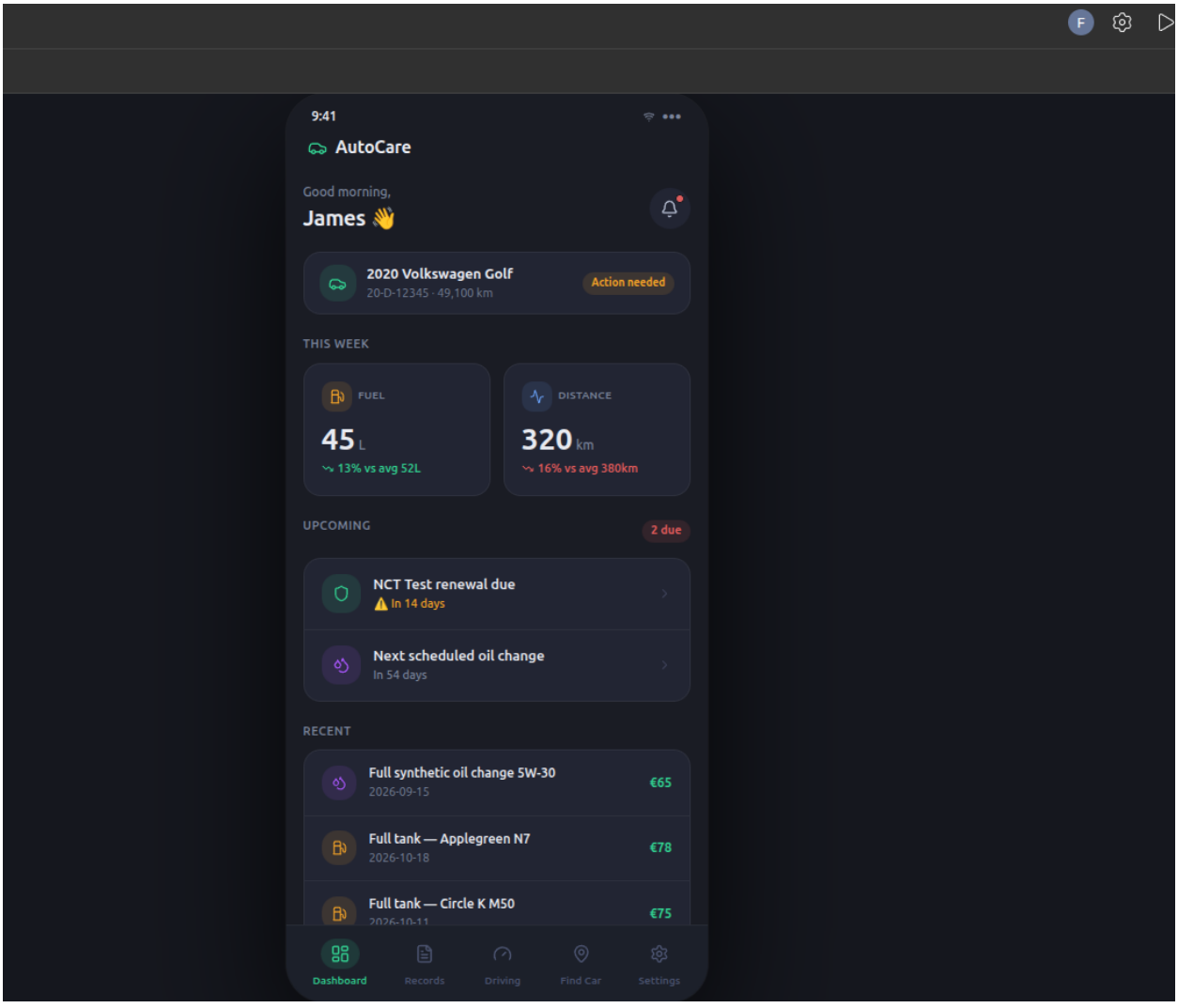
Android version	Andriod 13 and above
Internet connection	Required for online maps and navigation/ app can still be used offline

2. STRUCTURE:

The first build phase of the application will be offline (no backend yet, local storage), this to reduce complexity when developing the database rather than using a cloud service from the biggening. Once the front side is finalized and database is almost complete, I will start building the server side of the application.

Figma illustration of the fisrt screen that app should show(after users intial setup):

/not the final design

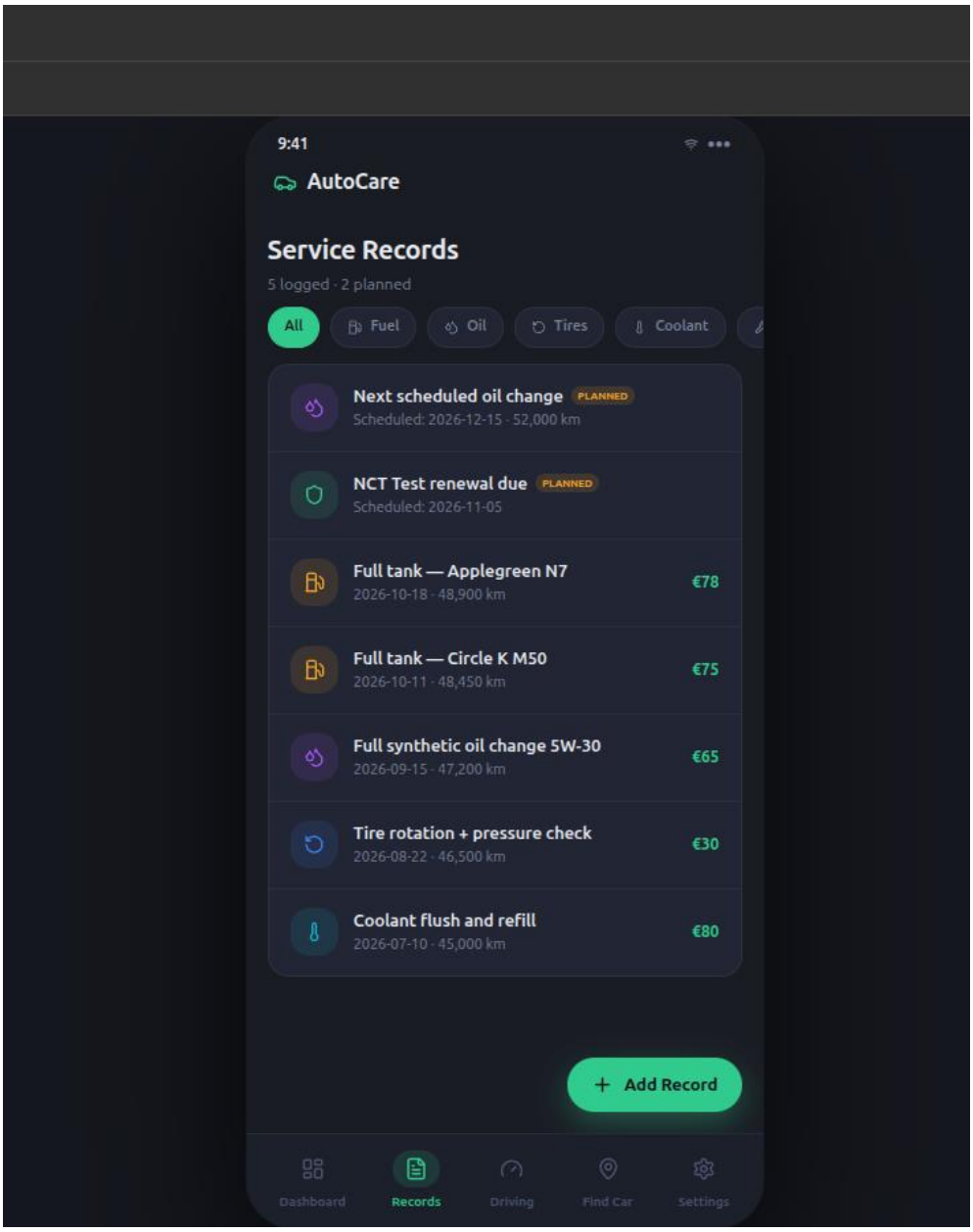


Dashboard: top card showing some details about the vehicle, bollow it two side by side cards: first showing fuel spending for the current week and compared to average (will need at least one week of usage to start calculating average), sencond card showing distance travelled for the current week compared to the average distance travled weekly(same case). Right bellow, a alert card that showing due soon maintances that are logged previously by the user.
The bottom panel contains(dashboard, records, driving mode, find my car, settings)

Records: contains these categories(fuel, oil, tyres, coolant, engine, electrical, NCT)

Each catogary has two functions: add racord, plan maintenance

What each catogory has not been finilized yet but here is an example: (Fuel: Litres, price, odometer, full tank | Engine: Repairs, inspections, parts | NCT: Test date, result, expiry)



Driving mode: A full screen map. Here we will use Google maps SDK. Embedded map, live position, GPS speedometer, journey distance and accelerometer monitoring.
(have not been detailed yet, still being considered depending on the complexity of API integration)

Find my care:

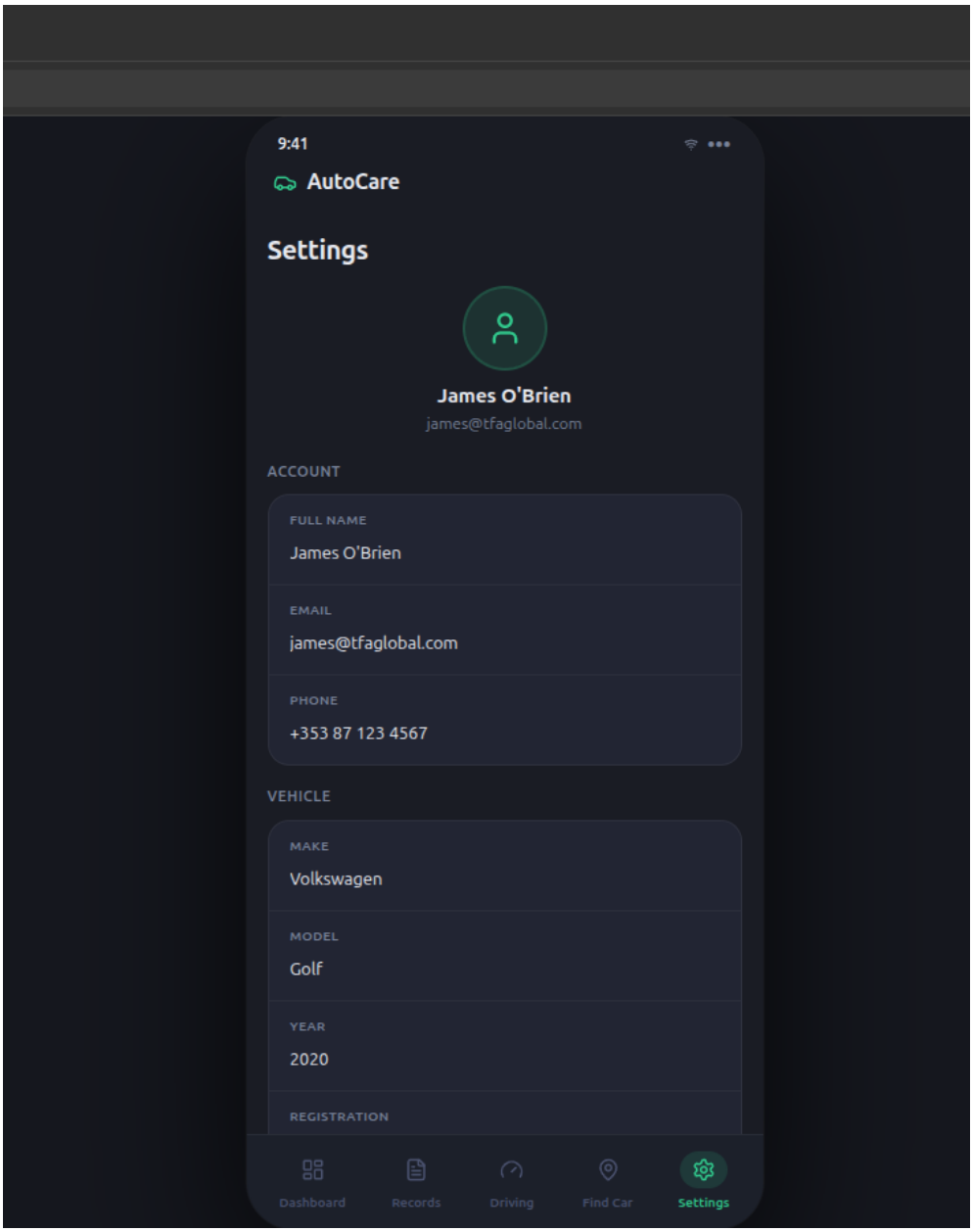
The user taps Save Parking Location when parked. AutoCare gets the current location, saves coordinates and time to Room, and displays a map marker.

Later, the user can return using the map, parking compass or external directions.

Reliable, straightforward and requires no continuous background detection.

On a later stage I'm considering using an API to automate the process, such saving coordinates when the Bluetooth is disconected

Settings:



Here where the user can edit his details